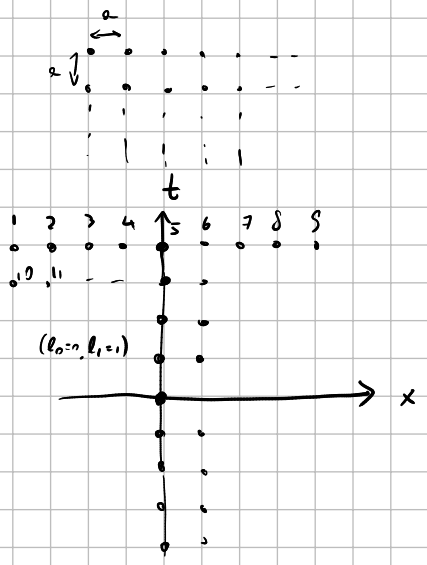
Look at field on a lattice: $\Lambda \subseteq \mathbb{R}^4$ discrete



a = lattice spacing

$$\Lambda = \left\{ \sum_{\mu=0}^{3} a\, e_\mu\, l_\mu \;\middle|\; l_\mu = -4, 4 \right\}$$

basis vector $(0, \dots, 1, \dots 0)$ at pos μ

$\phi : \Lambda \to \mathbb{R}$ field on a lattice

image of field: $\{ \phi(x) \mid x \in \Lambda \}$ discrete

if we label all the points in $\Lambda = \{ x_i \mid i = 1, \dots, N \}$

and we call $\phi_i = \phi(x_i)$ then a field is fully specified by $(\phi_1, \dots, \phi_N) \in \mathbb{R}^N$

$S : \mathbb{R}^n \to \mathbb{R}$

path integral on lattice $\displaystyle\int d^N\phi \; e^{-S(\phi)}$

N cardinality of lattice (number of points)

$$Z_0 = \int d^N\phi \; e^{-S_0(\phi)}$$

$\phi \in \mathbb{R}^N$

choose $S_0(\phi) = \frac{1}{2} \phi^t A \phi \in \mathbb{R}$ $\quad A$ symmetric positive definite matrix

without interactions

$$= \sum_{i,j} \frac{1}{2} \phi_i A_{ij} \phi_j$$

summing over $x \in \Lambda$

$$Z_0 = \sqrt{\det(2\pi A^{-1})}$$

$$\langle \phi_{i_1} \dots \phi_{i_n} \rangle_0 := \frac{1}{Z_0} \int d^N\phi \; e^{-S_0(\phi)} \phi_{i_1} \phi_{i_2} \dots \phi_{i_n}$$

Gaussian expectation value

$$\langle \phi_i \phi_j \rangle_0 = (A^{-1})_{ij}$$

Wick's theorem or Isserlis' theorem

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle_0 = \sum_{p \in \underbrace{P^2(i_1, \dots, i_n)}} \prod_{(i,j) \in p} \langle \phi_i \phi_j \rangle_0$$

$\phi_{i_1} \equiv \phi(x_{i_1}) \in \mathbb{R}$

↳ set of all partitions of $(i_1, i_2, \dots, i_n)$ into pairs

$$\langle \phi_{i_1} \phi_{i_2} \phi_{i_3} \phi_{i_4} \rangle_0 = \langle \phi_{i_1} \phi_{i_2} \rangle_0 \langle \phi_{i_3} \phi_{i_4} \rangle_0 + \langle \phi_{i_1} \phi_{i_3} \rangle_0 \langle \phi_{i_2} \phi_{i_4} \rangle_0 + \langle \phi_{i_1} \phi_{i_4} \rangle_0 \langle \phi_{i_2} \phi_{i_3} \rangle_0$$

$$\langle \phi_1 \phi_1 \rangle = (A^{-1})_{11}$$

$$\langle \phi_1 \phi_2 \rangle = (A^{-1})_{12}$$

<u>Consequence:</u> $\langle \phi_{i_1} \cdots \phi_{i_n} \rangle_0 = 0$ if n is odd $\quad$ ($\langle \phi_i \rangle = 0$, $\langle \phi_i \phi_j \phi_k \rangle = 0$ etc.)

Now we add interactions $\quad S(\phi) = S_0(\phi) + S_I(\phi)$

<u>ex:</u> $\quad S_I(\phi) = g \sum_i \phi_i^4 \qquad \phi^4$ theory

↳ real parameter (strength of interaction)

partition function

$$Z(g) = \int d^N\phi \; e^{-S(\phi)} \qquad S(\phi) = S_0(\phi) + S_I(\phi)$$

expand w.r.t. to g

$$e^{-g \sum_i \phi_i^4} = \sum_{k=0}^{\infty} \frac{(-g)^k}{k!} \left(\sum_i \phi_i^4 \right)^k$$

$$Z(g) \sim \sum_{k=0}^{\infty} \frac{(-g)^k}{k!} \int d^N\phi \; e^{-S_0(\phi)} \left(\sum_i \phi_i^4 \right)^k \quad \text{asymptotic}$$

$$= \sum_{k=0}^{\infty} \frac{(-g)^k}{k!} Z(0) \left\langle \left(\sum_i \phi_i^4 \right)^k \right\rangle_0$$

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle_0 = \frac{1}{Z(0)} \int d^N\phi \; e^{-S_0(\phi)} \phi_{i_1} \cdots$$

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle \sim \frac{1}{Z(g)} \int d^N\phi \; e^{-S(\phi)} \phi_{i_1} \cdots \phi_{i_n}$$

amplitudes in physics

$$= \frac{1}{Z(g)} \sum_{k=0}^{\infty} \frac{(-g)^k}{k!} \int d^N\phi \; e^{-S_0(\phi)} \left(\sum_i \phi_i^4 \right)^k \phi_{i_1} \cdots \phi_{i_n}$$

$$= \frac{Z(0)}{Z(g)} \sum_{k=0}^{\infty} \frac{(-g)^k}{k!} \langle \phi_{i_1} \cdots \phi_{i_n} \left(\sum_i \phi_i^4\right)^k \rangle_0$$

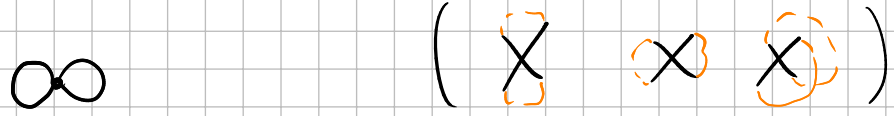
$$Z(g) \sim Z(0) - g\, Z(0) \langle \sum_i \phi_i^4 \rangle_0 + \frac{g^2}{2} Z(0) \langle \sum_i \phi_i^4 \sum_j \phi_j^4 \rangle_0$$

Feynman diagrams

- for each interaction term $\sum_i \phi_i^4$ we drew a vertex with 4 legs (internal legs because we sum over indices)

- because we are taking the Gaussian average, we need to "contract" legs attach them to each other in pairs

$$\sum_i \langle \phi_i \phi_i \phi_i \phi_i \rangle_0 = \sum_i 3 \langle \phi_i \phi_i \rangle_0 \langle \phi_i \phi_i \rangle_0$$



$\hookrightarrow -g \sum_i \langle \phi_i \phi_i \rangle_0 \langle \phi_i \phi_i \rangle_0$ $\qquad \langle \phi_i^4 \rangle_0$

- each vertex carries a factor of $-g$ and a sum over an internal index i (each leg of the vertex carries the label i)

- each connection between two legs is a propagator $\langle \phi_i \phi_j \rangle_0$

$= \langle \phi_i \phi_j \rangle_0$

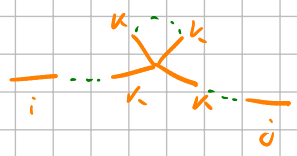
Now look at $\langle \phi_i \phi_j \rangle \approx \frac{Z(0)}{Z(g)} \left(\langle \phi_i \phi_j \rangle_0 - g \langle \phi_i \phi_j \sum_k \phi_k^4 \rangle_0 \right)$ (drawings)

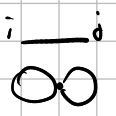
$-g \langle \phi_i \phi_j \sum_k \phi_k^4 \rangle_0$ how to obtain with drawings

1 interaction vertex and 2 external legs $\underline{\quad i \quad}$ $\underline{\quad j \quad}$

$\langle \cdots \rangle_0$ takes all possible contractions

① connect i to j

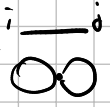


$$-g \langle \phi_i \phi_j \sum_k \phi_k^4 \rangle_0 = 3$$  $$+ 12$$

factorial only with odd numbers

the number of partitions into pairs of $2n$ elements is $(2n-1)!!$

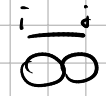
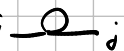
$2n = 6 \Rightarrow 15$

 $= -g \sum_k \langle \phi_k \phi_k \rangle_0^2 \langle \phi_i \phi_j \rangle_0$ (vacuum diagram)

$= -g \sum_k \langle \phi_i \phi_k \rangle_0 \langle \phi_j \phi_k \rangle_0 \langle \phi_k \phi_k \rangle_0$

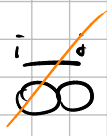
$Z(g) \sim Z(0)$ ($1 + 3$) $\qquad \frac{1}{Z(g)} \sim \frac{1}{Z(0)}$ ($1 - 3$)

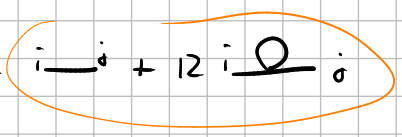
$$\frac{1}{1+g} \approx 1 - g$$

$\langle \phi_i \phi_j \rangle_0 - g \langle \phi_i \phi_j \sum_k \phi_k^4 \rangle_0 =$ $+ 3$  $+ 12$ 

$\frac{Z(0)}{Z(g)} \left(\langle \phi_i \phi_j \rangle_0 - g \langle \phi_i \phi_j \sum_k \phi_k^4 \rangle_0 \right)$

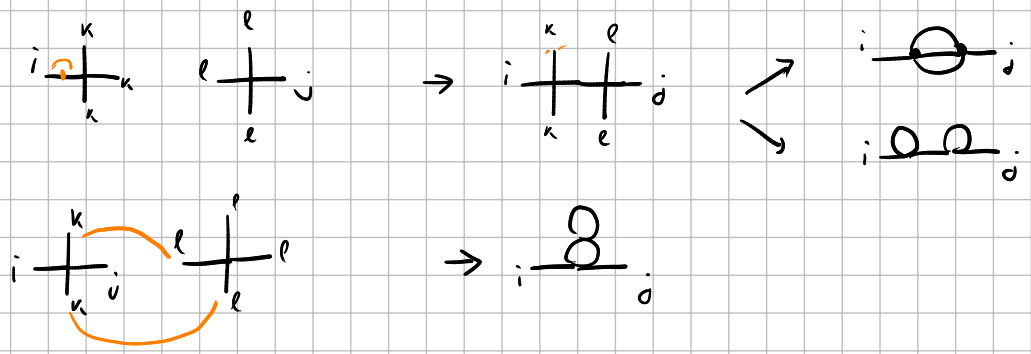
$=$ ($1 - 3$) ($+ 3$ $+ 12$)

$=$ $+ 3$  $+ 12$ $- 3$ $+ o(g^2)$

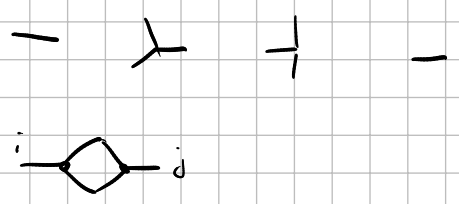
$=$ 

when we look at $\langle \phi_i \phi_j \rangle$ to order n the vacuum diagrams will drop

$\langle \phi_i \phi_j \rangle$ to second order



$$\langle \phi_i \phi_j \rangle = \quad + 12 \quad + \frac{1}{2!}\Big(A \quad + B \quad + C \quad \Big)$$

$$g^2 \sum_{k,l} \langle \phi_i \phi_k \rangle \langle \phi_j \phi_l \rangle \langle \phi_k \phi_l \rangle^3$$

$$g^2 \sum_{k,l} \langle \phi_i \phi_k \rangle \langle \phi_j \phi_l \rangle \langle \phi_k \phi_l \rangle \langle \phi_k \phi_k \rangle \langle \phi_l \phi_l \rangle$$